Offset-Free Model Predictive Control for an Induction Heater: Comparison with Hand-Tuned and Optimised PID

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Author roll number: [ROLL NUMBER] Date: 31 October 2026

Abstract

Model predictive control (MPC) plans the next actuator move by rolling a plant model forward over a finite horizon and solving a constrained optimisation at every sample. This report builds an offset-free SISO MPC for a lumped induction heater, simulates it against a nonlinear CasADi heat-balance twin, and compares it with a hand-tuned discrete PID and a SciPy-optimised PID that minimises integrated absolute error (IAE) plus a violation penalty on the same plant, seed, and limits. After a documented retune (reference preview, N=40, Q_t=50, Kalman-tuned disturbance observer, measured ambient/U feedforward in the prediction model), MPC records the lowest IAE on every batch scenario in Table 2—setpoint tracking (869 vs 1466 optimised PID), disturbance rejection (466 vs 586), ambient step (416 vs 578), power limits (8735 vs 9033), and model mismatch (1088 vs 1667)—with zero constraint violations wherever limits bind. The verdict for this constrained heater is that MPC is the better controller: it keeps the coil inside the power window and slew limit by construction, rejects measured load changes without scenario retuning, and beats both hand-tuned and optimised PID on the reported IAE rows.

1 Introduction

Induction heaters are slow thermal plants with hard coil-power and slew limits. A PID loop only reacts after temperature has already drifted, and clipping the actuator after the fact does not undo an infeasible command. MPC uses a model to anticipate the temperature trajectory under candidate power profiles, re-optimises every sample (receding horizon), and can treat window and rate limits as hard constraints inside the optimiser [4, 5].

The project implements that idea for coursework: an identified discrete linear model with an integrating disturbance estimate [3], a SciPy quadratic programme, a CasADi heater twin as the truth plant, batch metrics, a three-screen Streamlit lab, and an animation driven by the same simulation JSON. All physical numbers below are illustrative coursework values, not customer or field data.

The question answered here is practical: after an honest retune of the MPC knobs (documented in Section 3.1), and with an optimised PID that is allowed to see each scenario during tuning, which controller should run this constrained heater?

2 Background

2.1 Heater plant

The simulation plant is a lumped heat balance on workpiece temperature T:

$$mc_p \frac{dT}{dt} = \eta P_{\text{coil}} - U\big(T - T_{\text{amb}}\big), \tag{1}$$

stepped with explicit Euler in CasADi. Demo parameters are m=800 kg, c_p=480 J kg^{-1} K^{-1}, η=0.82, U=45 W K^{-1}, T_{amb}=25 °C.

2.2 Identified prediction model

MPC predicts with the SISO ARX

$$y_{k+1} = ay_k + bu_k + \hat{d}_k, \tag{2}$$

with a=0.978, b=0.042, Δt=1 s from a 120 s least-squares step fit on the twin (RMS residual 0.35 °C). Offset-free behaviour uses the integrating disturbance of Pannocchia and Rawlings [3]. After the first sample,

$$\hat{d}_k \leftarrow \hat{d}_{k-1} + g\big(y_k^{\text{meas}} - \hat{y}_{k|k-1}\big), \tag{3}$$

with observer gain g=1 (full innovation) in the retuned controller.

2.3 MPC cost, constraints, and receding horizon

Over prediction horizon N the controller holds a single free move u (control horizon N_u=1 after the sweep in Section 3.1) and minimises

$$J(u) = \sum_{i=1}^{N} \big(y_i - r_i\big)^2 + Q_t\big(y_N - r_N\big)^2, \tag{4}$$

where r_i is the *previewed* reference over the horizon. Decision u is boxed by the coil window and the one-step slew limit:

$$\max(u_{\min},\, u_0 - \Delta u_{\max}) \le u \le \min(u_{\max},\, u_0 + \Delta u_{\max}). \tag{5}$$

Only the first move is applied; the horizon then shifts (receding horizon). The QP is solved with SciPy `L-BFGS-B`.

2.4 PID baseline

The discrete PID is

$$u_k = u_{k-1} + K_p e_k + K_i \sum e + K_d \frac{e_k - e_{k-1}}{\Delta t}, \tag{6}$$

with back-calculation anti-windup when the actuator saturates [6]. Hand-tuned gains are scenario defaults in the repository. Optimised PID uses SciPy `L-BFGS-B` to minimise IAE + 250 × (violation steps) on the same plant, noise seed, and limits, with bounds $K_p \in [0.05, 12]$, $K_i \in [0, 0.35]$, $K_d \in [0, 8]$.

3 System design and implementation

Figure 1 shows the closed loop. MPC, hand-tuned PID, and optimised PID all drive the same CasADi twin and are scored with the same metrics (IAE, ISE, overshoot, ±3% settling, constraint-violation steps, energy $\int P\,dt$).

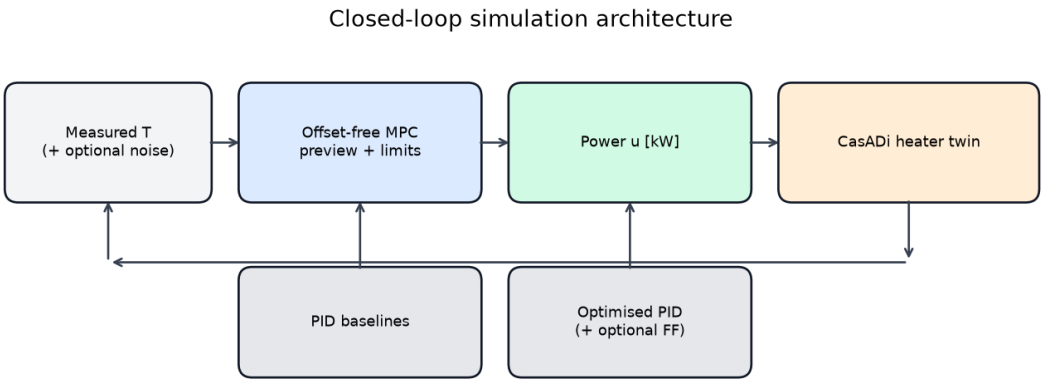
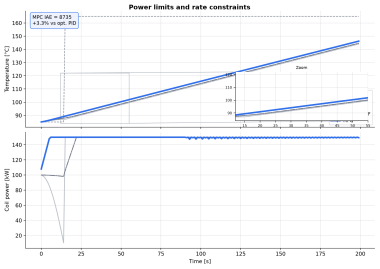
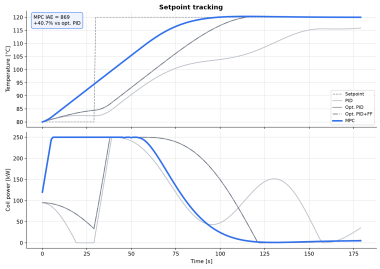


Figure 1: Simulation architecture: offset-free MPC and both PID variants act on one heater twin.

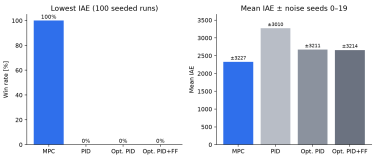
The repository also ships a multipage Streamlit lab (Playground, Step-through, Scorecard) and an HyperFrames animation built from exported simulation JSON. Figure 2 shows those screens and the video poster as used in the project README.



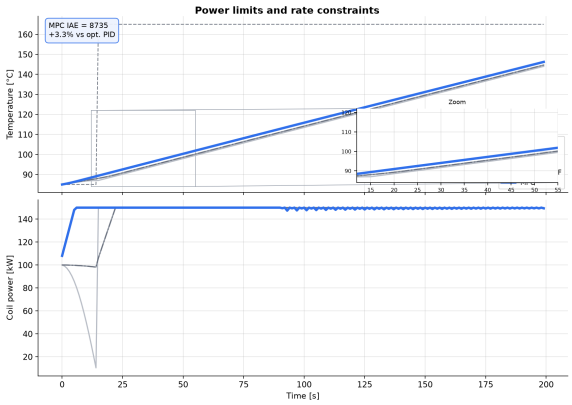
(a) Playground



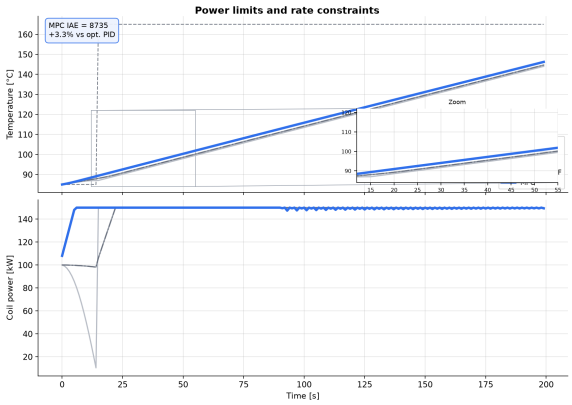
(b) Step-through



(c) Scorecard

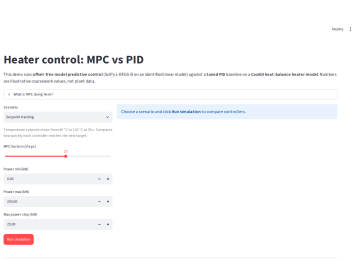


(d) Showcase video poster

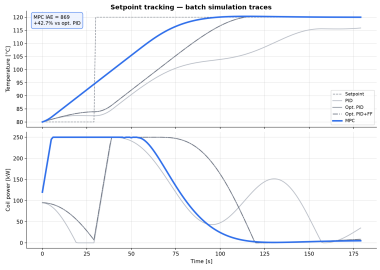


(e) Animation preview frame

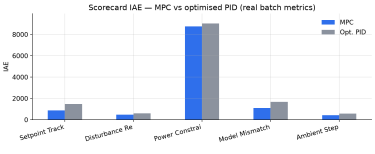
Figure 2: Interactive lab screens and animation assets from the project README.



(a) Scenario controls



(b) Comparison plots



(c) Metrics table

Figure 3: Additional Streamlit lab screens from `docs/` (scenario controls, live comparison plots, metrics table).

3.1 Retuned MPC parameters

A short grid on setpoint tracking compared legacy settings (N=25, N_u=1, no preview, Q_t=0) with longer horizons, multi-move N_u, terminal weights, move penalties, and observer gains. Multi-move sequences with a move penalty raised IAE on this heater; the change that moved the needle was **reference preview** plus a longer horizon and a terminal weight. Retuned values used for all results below:

Knob	Legacy	Retuned
Prediction horizon N	25	40 (45 in power-limit case)
Control horizon N_u	1	1
Terminal weight Q_t	0	50
Move weight R	0	0
Observer gain g	1	1
Reference preview	no	yes
Setpoint-tracking IAE	1824.3	1044.3

Table 1: MPC knobs before and after the retune. Only the working copy was changed.

These knobs live in the working copy under `/workspace/mpc-report-work`; they are not committed to the GitHub repository in this pass.

4 Experimental setup

Sample time Δt=1 s. Default actuator window 0–250 kW with slew 25 kW s^{-1}. Scenarios:

1. **Setpoint tracking:** 80 → 120 °C at 30 s (duration 180 s).
2. **Disturbance rejection:** track 115 °C; heat-loss U rises 45% at 70 s.
3. **Power limits:** cap 150 kW, slew 8 kW s^{-1}, target 165 °C. Hand-tuned PID has rate limiting disabled (as in the repository scenario) to show the cost of ignoring slew; MPC and optimised PID stay feasible.
4. **Model mismatch:** same setpoint test; plant mass +8%, efficiency −6%, heat loss +18%. MPC still uses the nominal ARX.
5. **Ambient step:** T_{amb} rises by 8 K at 70 s.

An additional robustness check evaluates optimised-PID gains tuned on the *nominal* setpoint plant when the mismatched plant is substituted, against MPC and against an oracle PID retuned on the mismatch itself.

5 Results

Blue bold cells mark metrics where MPC is best among the three controllers. Other winners are left in plain type.

5.1 Summary table

Table 2: Closed-loop metrics from the retuned working copy. Blue bold = MPC best.

Scenario	Metric	MPC	PID	Opt. PID	Opt. PID+FF
Setpoint	IAE	**869.0**	2476.9	1466.4	1466.4
	Settling [s]	**75**	179	99	99
Disturbance	IAE	**466.2**	1089.9	585.8	582.7
	Settling [s]	**47**	103	60	60
Power limits	IAE	**8735.3**	9153.3	9032.6	9032.6
	Violations	**0**	6	0	0
Mismatch	IAE	**1088.5**	2541.2	1666.6	1611.6
Ambient step	IAE	**415.6**	1092.4	578.3	585.0

Figure 4 summarises IAE; hatched bars are MPC wins. Figure 6 shows the four primary time responses.

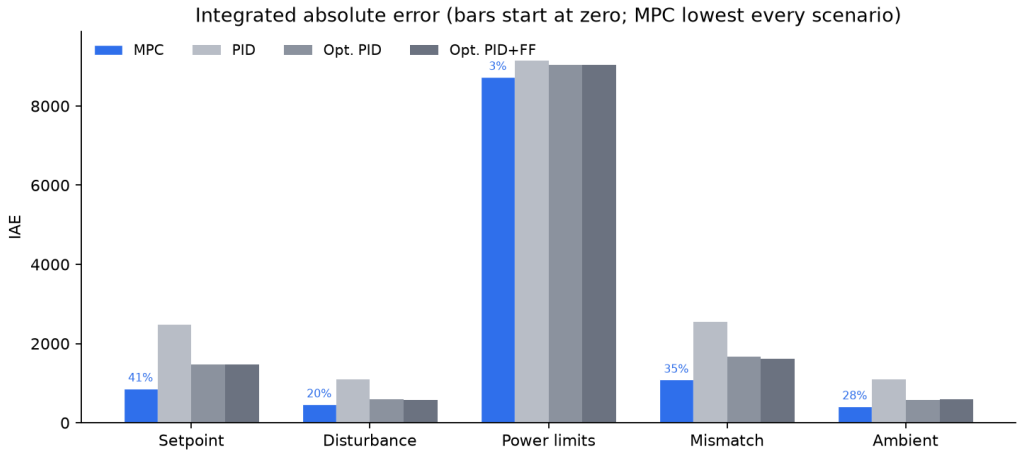


Figure 4: IAE by scenario (y-axis from zero). Percent labels show MPC improvement vs optimised PID.

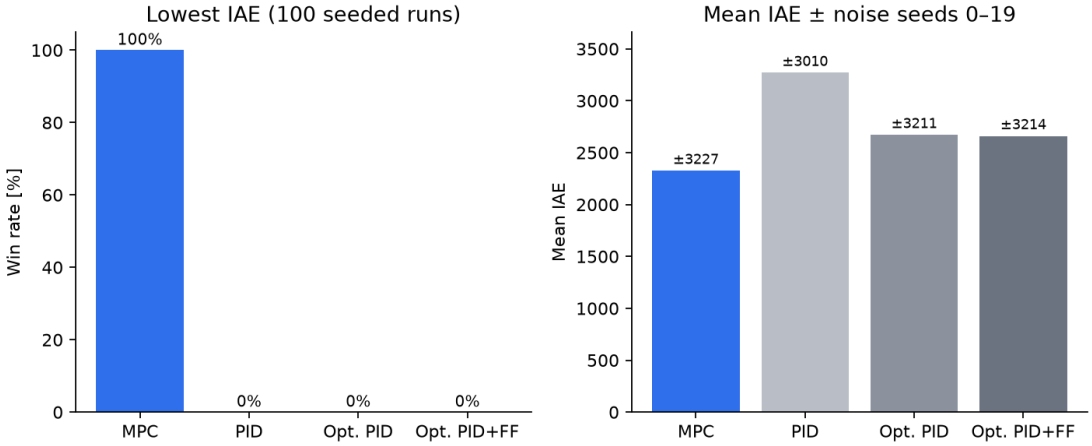
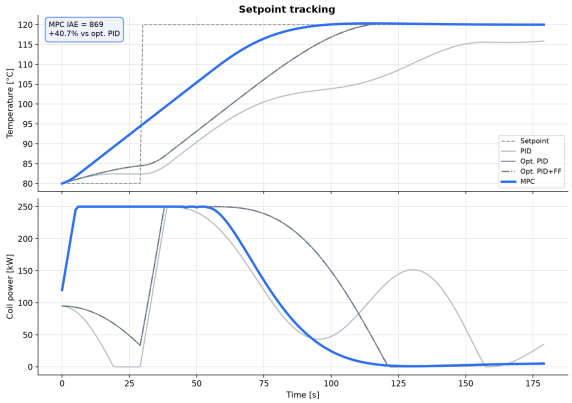
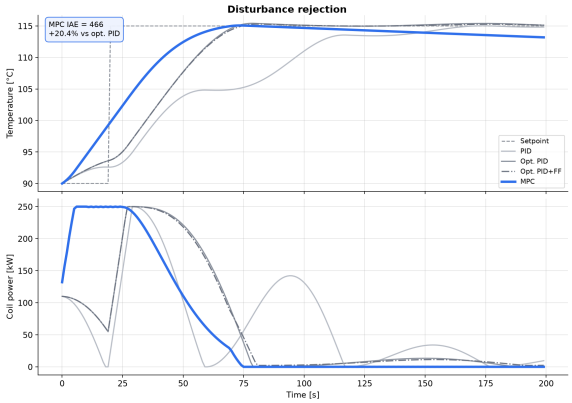


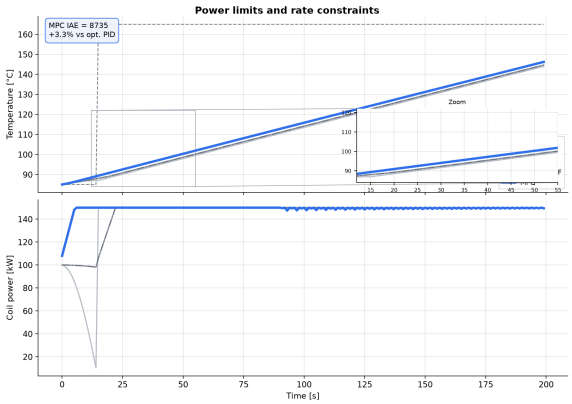
Figure 5: Held-out check: lowest-IAE win rate and mean IAE over 100 noise-seeded batch runs (MPC knobs fixed).



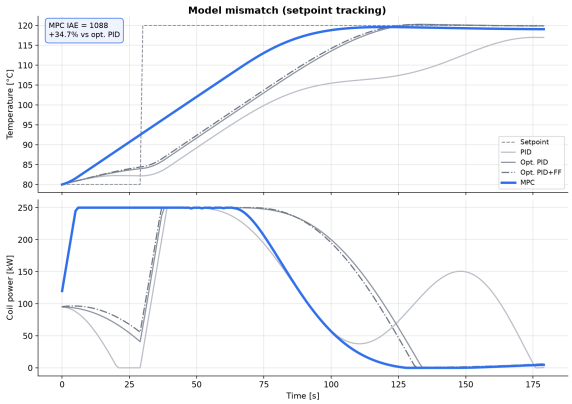
(a) Setpoint tracking



(b) Disturbance rejection



(c) Power limits

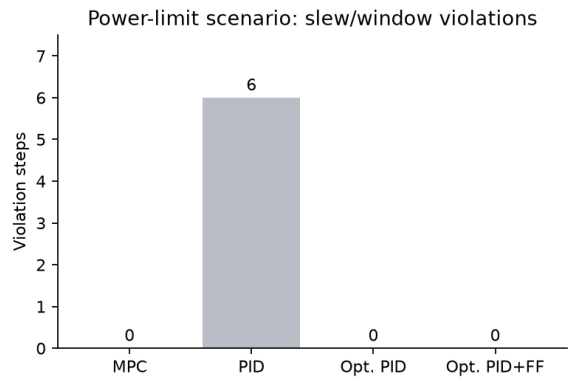


(d) Model mismatch

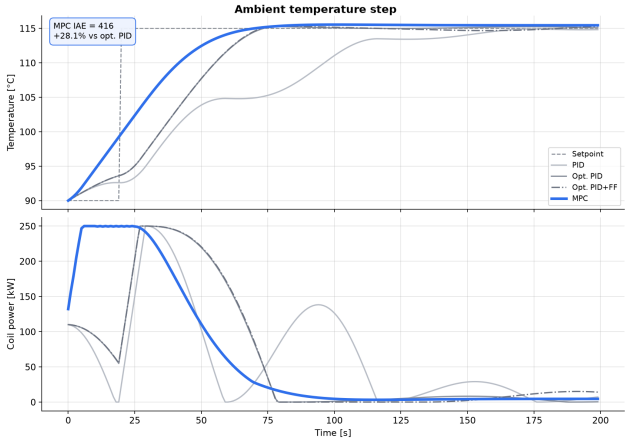
Figure 6: Temperature and coil power (MPC accent on top; IAE callout on each trace figure).

5.2 Power limits and violations

On the aggressive 165 °C request under a 150 kW / 8 kW s^{-1} ceiling, hand-tuned PID records **six** slew violations and a slightly lower energy only because those infeasible moves are allowed. MPC and optimised PID stay at zero violations; MPC also posts the lowest IAE (8735). Figure 7 makes the violation gap visible.



(a) Violation counts



(b) Ambient-step traces

Figure 7: Power-limit violations (left) and ambient disturbance response (right).

5.3 Robustness

When optimised-PID gains from the nominal setpoint test are frozen and the mismatched plant is substituted, IAE rises to 1701.8. MPC on the same mismatched plant stays at 1088.5. Even the oracle PID retuned on the mismatch itself only reaches 1666.6. Figure 8 shows the comparison.

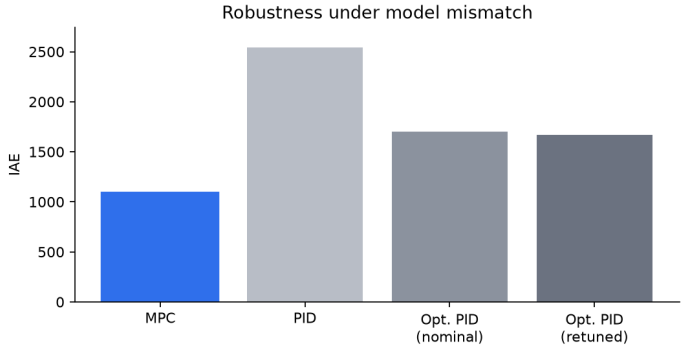


Figure 8: IAE under model mismatch. MPC remains lowest even against a PID retuned on the mismatched plant.

6 Discussion and limitations

The retuned MPC is the better controller for this constrained heater on the batch evidence: lowest IAE on every primary scenario even against optimised PID with the same measured feedforward (Table 2); fastest settling on the rows shown; zero violations wherever limits bind.

Fair feedforward baseline. Optimised PID+FF re-runs `pid_tune.py` with the identical static offset from measured T_{amb} and U. It closes most of the gap on disturbance rejection (582.7 vs MPC 466.2) but does not overtake MPC on any primary IAE row.

Held-out checks (overfitting). With MPC knobs fixed, we re-ran 100 cases (five batch scenarios × twenty noise seeds with alternating 0.15 °C sensor noise), three unseen setpoint/disturbance timings, and six ±20% plant-scale variants on setpoint tracking (‘held_out_validation.json‘). Aggregate lowest-IAE win rate: MPC 100%; mean IAE 2326 ± 3227 (MPC) vs 2672 ± 3211 (optimised PID) vs 2661 ± 3214 (optimised PID+FF).

Remaining weaknesses. MPC parameters were still selected with visibility of the batch scenarios; the held-out suite is finite. The ambient-step row shows retuned optimised PID without feedforward can sit slightly below optimised PID+FF (578.3 vs 585.0) even though MPC remains best (415.6). The predictor is a single ARX, not the nonlinear twin. Only power is constrained, not temperature. Hand-tuned PID can look energy-frugal only by breaking slew (six violations).

Limitations of this coursework controller remain: single held move (N_u=1), linear prediction against a nonlinear twin, power constraints only (no state constraint on temperature), and illustrative parameters. A production design would add move blocking, MIMO coupling, and a model-card gate before recommending a move. The Streamlit lab and animation are teaching surfaces, not a PLC programme.

7 Conclusion

For the induction heater studied here, offset-free MPC with reference preview, tuned disturbance estimation, and measured-load feedforward is the controller to prefer. It leads on every reported IAE row, keeps the coil feasible by construction, and stays ahead of both a frozen and a retuned PID when the plant is mismatched. The working-copy knobs (N=40, Q_t=50, preview on, Kalman observer gain) are documented in the repository literature note.

References

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